

electron/code-city-model.cjs

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Overview

`electron/code-city-model.cjs` is a new CommonJS module that builds a Code-City model payload from two project outlines. It adds utilities for normalizing side names, determining dominant authors, mapping blame data, and assembling file and symbol nodes.

Key changes

- `normalizeSide` (lines 5-7) – normalizes `"baseline"` or `"target"` strings, defaulting to `"target"`.
- `dominantAuthor` (lines 9-11) – returns the first author in an owners array or `null`.
- `ownershipForRange` (lines 13-35) – counts author line-counts for a given range and returns up to six top owners.
- `selectBlamePaths` (lines 38-54) – sorts file paths by change kind and size, then limits to 140 entries.
- `buildBlameMap` (lines 57-62) – creates a map of relative paths to blame ownership using `blameFileOwnership`.
- `buildSymbolFileIndex` (lines 65-77) – indexes symbol spans by file path.
- `buildSideNodes` (lines 79-139) – constructs file and symbol node arrays for a side, aggregating authors and dominant authors.
- `buildCodeCityModelPayload` (lines 142-174) – orchestrates outline loading, blame mapping, side node construction, and returns a payload with `generatedAt`, labels, files, symbols, and sorted authors.
- **Import** – `const { blameFileOwnership } = require("../git-blame.cjs");` (line 3).
- **Export** – `module.exports = { buildCodeCityModelPayload };` (line 176).

Impact

- **Correctness** – the module relies on well-formed outlines; missing `symbol_spans_by_file` entries are silently ignored.
- **Maintainability** – side-node logic is centralized, easing future blame-handling changes.
- **Performance** – the implementation iterates over all files and symbols; large projects may incur higher runtime and memory usage.
- **Compatibility** – uses CommonJS, so existing Node environments remain unaffected.

Risks & follow-ups

- **Testing** – no unit tests currently cover this module; adding tests for `ownershipForRange` and `buildCodeCityModelPayload` is advisable.
- **Error handling** – functions assume valid input; behavior with malformed outlines is undefined.

- **Blame data size** – `buildBlameMap` loads blame for each selected path; monitor memory usage on very large repositories.
- **Author selection** – `dominantAuthor` simply picks the first owner; confirm this aligns with business expectations.